

CLIFFE OWINO IBANDE

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Professional Summary

Results-driven Associate Software Engineer with a Bachelor's in Software Engineering and hands-on experience in DevOps, full-stack development, AI/ML, and cloud architectures. Proficient in building scalable microservices, automating CI/CD pipelines, and deploying AI-driven applications using tools like Docker, Kubernetes, and Hugging Face. Experienced in supporting high-availability systems for M-Pesa platforms across multiple African countries through Prodapt's award-winning partnership with Vodacom, ensuring operational excellence for one of Africa's largest mobile money services. Passionate about leveraging emerging technologies like large language models to solve real-world problems and drive business efficiency. Seeking opportunities to contribute to innovative tech teams.

Education

Kisii University, Kisii County, Kenya

Bachelor of Science in Software Engineering

2020 – Dec 2024

Maseno School, Kisumu County, Kenya

Kenya Certificate of Secondary Education

2016 – Dec 2020

Certifications & Additional Courses

- Cloud Architecture & DevOps Essentials – AWS/Azure (2023)
- SPSS for Data Analysis (2024)
- Huawei Cloud Fundamentals (2023)
- Docker Fundamentals (2025)
- Postman API Development (2025)
- Computer Architecture and Design (2026)

Technical Skills

Languages: Python, JavaScript, Java, TypeScript, HTML, CSS, SQL, Bash

Web & Mobile Frameworks: Node.js, React, Frappe, Next.js, Django, Flask, React Native, Spring Boot, Material-UI, GraphQL

Developer Tools: Git, Docker, Kubernetes, Jenkins, VS Code, PyCharm, IntelliJ, Eclipse, Linux (Ubuntu/CentOS), Cron jobs, Shell Scripting, NGINX, Virtualization (VMware, VirtualBox), Terraform, Ansible

Libraries & AI: Pandas, NumPy, Matplotlib, Seaborn, scikit-learn, TensorFlow, PyTorch, LangChain, Hugging Face Transformers, OpenAI, NLTK, spaCy

Cloud & DevOps: Huawei Cloud, AWS, Azure, Google Cloud Platform, Microservices, CI/CD pipelines, REST & gRPC services, Grafana, Prometheus, Loki (Monitoring & Logging), ELK Stack

Databases: MongoDB, PostgreSQL, MySQL, Redis

Data Analysis Tools: SPSS, Excel Analysis, Python Scientific Stack

Experience

Associate Software Engineer

Prodapt Ltd

September 2025 – Present

Supporting M-Pesa operations across 5 nations: Mozambique (MZ), Lesotho (LS), Tanzania (TZ), Ghana (GH), and Democratic Republic of Congo (DRC) through Prodapt's award-winning partnership with Vodacom for M-Pesa Africa. Working under the Core Payment Services (CPS) division, ensuring reliability, availability, and security of mobile money platforms that process billions in transactions and serve millions of users across Africa.

- Managed and maintained large-scale Linux-based production servers supporting M-Pesa's core payment infrastructure, handling over 1M daily transactions.
- Worked extensively with the Huawei technology stack, including cloud services, monitoring tools, middleware, and network components.
- Performed database monitoring, tuning, troubleshooting, and optimized queries for high-volume transactional systems, reducing latency by 15%.

- Handled VM provisioning, virtualization management, and performance tracking across distributed infrastructures using tools like VMware and Kubernetes.
- Executed certificate creation, renewal, and deployment for secure communication between payment services, ensuring 99.9% uptime.
- Diagnosed and resolved network-related issues, including latency, routing, connectivity, and service availability problems.
- Conducted data analysis and reporting to identify system bottlenecks, anomalies, and operational irregularities using Python and Grafana.
- Collaborated with cross-functional teams to support incident management, service restoration, and root-cause analysis.
- Utilized monitoring tools (Grafana, Digital View) to ensure system health and SLA compliance.
- Automated repetitive tasks and wrote custom automation scripts using Python, Bash, and internal tools, improving efficiency and reducing manual errors by 30%.
- Contributed to the upcoming rollout of TZ Fintech 2.0, a cloud-based platform upgrade enhancing scalability, AI-driven fraud detection, and transaction capacity for Tanzania's M-Pesa services.

Backend Web Developer

Techsavannah

Sep 2024 – Jul 2025

Delivered bespoke software solutions and IT consulting services for diverse industry clients.

- Integrated microservices architectures, Dockerized applications, and Kubernetes deployments for scalable cloud environments, supporting 10+ client projects.
- Implemented Linux-based server automation, cron jobs, monitoring dashboards, and deployment pipelines using Jenkins and Terraform.
- Collaborated with cross-functional teams to optimize business processes and modernize legacy systems.
- Developed high-availability systems with API gateways, reverse proxies, and load balancing (NGINX), achieving 99% availability.

Undergraduate Industrial Attachment (Data Analyst)

Huawei Tech. (Kenya) CO., LTD

June 2023 – Aug 2023

- Designed and refined data models for efficient data organization and management.
- Built interactive dashboards improving visualization and strategic decision-making using Python and Excel.
- Assisted in constructing and maintaining data cubes for analytics.
- Generated advanced network monitoring reports and performance insights.
- Conducted data analysis using SPSS, Python, and Excel, identifying trends in network performance.
- Gained exposure to cloud architecture, virtualization, cloud monitoring, and distributed systems.

Projects

Fullstack & DevOps Projects

Linux-Based Monitor & Microservice Deployment

October 2025 – Present

Linux, Docker, Kubernetes, Prometheus, Grafana, Microservices (FastAPI/Node.js), Terraform

- Built a production-ready microservice application deployed on Docker containers, featuring services for user authentication, data processing, and API endpoints.
- Orchestrated services using Kubernetes with horizontal scaling and health checks, enabling auto-scaling based on CPU usage thresholds to handle variable loads.
- Implemented monitoring using Prometheus + Grafana with custom metrics and ELK Stack for logging, providing real-time dashboards for CPU, memory, and request latency.
- Automated deployments using shell scripts and cron jobs, reducing deployment time by 40% and minimizing downtime during updates.

- Integrated alerting mechanisms to notify via email/Slack for critical thresholds, improving system reliability in simulated production environments.

RAG-Based Document Intelligence System

September 2025 – October 2025

Python, LangChain, Hugging Face, FastAPI, MongoDB, Docker

- Built a Retrieval-Augmented Generation (RAG) pipeline for intelligent search and document processing, capable of handling PDF, DOCX, and text files up to 100MB.
- Integrated vector databases for high-speed semantic search, using embeddings from Hugging Face models to achieve 95% recall on test queries.
- Deployed scalable inference pipelines using Docker, supporting concurrent processing of multiple user requests without performance degradation.
- Used FastAPI to expose LLM endpoints with secure token auth, including rate limiting to prevent abuse and ensure data privacy.
- Tested with real-world datasets like legal documents and research papers, generating accurate summaries and answers with reduced hallucination.

NGINX SSL Automation & Certificate Management

Jan 2025 – Feb 2025

Linux, NGINX, OpenSSL, Certbot, Bash, Ansible

- Implemented automated SSL certificate creation and renewal using Certbot and cron, handling Let's Encrypt certificates for multiple domains.
- Configured NGINX reverse proxy for secure microservices traffic routing, including HTTPS redirection and header security policies like HSTS.
- Developed Ansible playbooks for consistent deployment across servers, reducing setup time from hours to minutes.
- Integrated logging and error handling to monitor certificate status, preventing expiration-related outages in test environments.

Fullstack Web Development Motimarket – Car Sales Platform

October 2025 – Present

React (TypeScript), Spring Boot REST Framework, PostgreSQL

- Developed a full-featured car sales web application with comprehensive listings and advanced search filters, supporting filters by make, model, year, price, and location.
- Implemented secure user authentication and efficient data management using Spring REST Framework and PostgreSQL, including JWT-based sessions and role-based access for buyers/sellers.
- Focused on a modern UI/UX design to enhance user engagement and conversion rates, incorporating responsive layouts and interactive elements like image galleries and live chat.
- Integrated third-party APIs for payment processing (e.g., Stripe simulation) and vehicle data lookup, handling over 1,000 sample listings with real-time updates.
- Optimized database queries and caching with Redis to reduce page load times by 25%, improving overall user experience.

The Board FC

August 2025 – September 2025

React (TypeScript), Supabase, Tanstack Query, Email Integration

- Built a fully functional modern sports web app featuring merchandise store, admin dashboard, blog, and fan zone, tailored for a football club with user profiles and match schedules.
- Implemented real-time data fetching with Tanstack Query for efficient state management, reducing API calls by 30% through caching.
- Integrated email notifications using services like SendGrid for updates on new blog posts, merchandise arrivals, and fan events.
- Designed admin dashboard with CRUD operations for content management, ensuring secure access with authentication layers.

Label Safi

Feb 2025 – August 2025

MERN Stack (MongoDB, Express.js, React, Node.js), Redux, Tailwind CSS

- Created a premium e-commerce platform for designer shoes and clothing with real-time inventory management and personalized recommendations based on user browsing history.

- Features a sophisticated shopping experience including wishlist, cart persistence, and seamless checkout process with integrated payment gateways.
- Utilized Redux for global state management and Tailwind CSS for responsive, customizable UI components across devices.
- Implemented search functionality with fuzzy matching and filters, handling a catalog of 500+ items with efficient MongoDB queries.
- Added admin panel for product uploads and order tracking, enhancing operational efficiency for store managers.

Royal Watches

August 2024 – September 2024

MERN Stack (MongoDB, Express.js, React, Node.js), Tailwind CSS, GraphQL

- Developed a dynamic e-commerce website featuring product listings, secure user authentication, and integrated payment gateways for luxury watches.
- Leveraged GraphQL for efficient data querying, allowing flexible client-side requests and reducing over-fetching of data.
- Used Tailwind CSS for a modern, responsive design with high-resolution image zoom and customizable watch configurators.
- Implemented user reviews and ratings system, with moderation tools to maintain content quality.

SkillUp

May 2024 – August 2024

Python, Flask, React, SQL, Docker, React Native, K-means, Git

- Created a machine learning-based web application that analyzes skill gaps to align workforce development, processing user resumes and job descriptions.
- Integrated modern RESTful APIs with React Native for seamless data visualization and collaboration tracking on mobile devices.
- Expanded functionalities to include personalized training recommendations using AI, leveraging K-means clustering for skill categorization.
- Dockerized the application for easy deployment, ensuring consistency across development and production environments.

- Conducted user testing with 50+ participants, refining the UI to improve recommendation accuracy by 20%.

Shutter Spot

Feb 2024 – Apr 2024

MERN Stack (MongoDB, Express.js, React, Node.js), Google Maps API, PostgreSQL, Socket.io

- Built a platform connecting photographers with a global clientele using location-based services, enabling real-time booking and portfolio sharing.
- Integrated Google Maps API to enhance location discovery and streamline client-photographer interactions, with geofencing for nearby matches.
- Delivered a user-friendly interface focused on simplicity and responsiveness, including live chat via Socket.io for instant communication.
- Implemented secure file uploads for photo portfolios and payment integration for service bookings.

AI, Machine Learning & LLM Projects

SkillUp AI Enhancement

June 2025 – August 2025

Python, Flask, React, SQL, Docker, K-means, Hugging Face Transformers, OpenAI

- Integrated large language models into the SkillUp platform to enhance natural language processing and data analysis, parsing resumes for skill extraction.
- Developed clustering algorithms and personalized recommendation engines to drive targeted training initiatives, using OpenAI APIs for semantic matching.
- Improved recommendation precision by fine-tuning models on domain-specific data, achieving 85% user satisfaction in beta tests.

ConversaAssist – Conversational AI Chatbot

April 2025 – May 2025

Python, Flask, Hugging Face Transformers, Docker

- Developed an intelligent conversational agent for customer support that leverages state-of-the-art LLM APIs, handling queries in multiple languages.

- Designed and implemented RESTful endpoints for real-time query handling and FAQ resolution, with context retention for multi-turn conversations.
- Integrated containerization for scalable deployment, reducing response times and enhancing user engagement by 35% in simulations.
- Incorporated sentiment analysis to route complex issues to human agents, improving resolution efficiency.

SmartDoc Summarizer – Automated Document Summarization Tool

Jan 2025 – March 2025

Python, FastAPI, Hugging Face Transformers, PostgreSQL, NLTK

- Built a web service that processes lengthy documents and generates concise summaries using pre-trained LLMs, supporting formats like PDF and Word.
- Implemented API endpoints to handle document uploads and summary retrieval, streamlining information consumption for users.
- Optimized summarization accuracy and performance through iterative testing and model fine-tuning, achieving 90% coherence in evaluations.
- Stored summaries in PostgreSQL for quick retrieval and version history, with NLTK for additional text preprocessing.

Cloud Projects

Huawei Cloud Resource Automation

Huawei Cloud ECS, OBS, Cloud Monitoring, Bash, Python

- Automated cloud resource creation and monitoring alerts using scripts to provision ECS instances and set up OBS storage buckets.
- Integrated container deployments on ECS instances, with automated scaling rules based on monitoring metrics.
- Developed alerting system to notify on resource thresholds, preventing over-provisioning and reducing costs by 20% in test scenarios.

Mobile Development

JamiiBudget

Sept 2024 – Jan 2025

React Native, Appwrite, TypeScript

- Developed a comprehensive full-stack expense tracking mobile application with robust financial management features, including budget categories and reports.
- Implemented secure user authentication and real-time analytics with Appwrite as the backend service, syncing data across devices.
- Enhanced user experience through intuitive interface design and responsive layouts, with charts for expense visualization.
- Added features like recurring expense reminders and export to CSV, tested on iOS and Android for cross-platform compatibility.

Additional Information

- **Networking & AI Enthusiast:** Continuously upgrading skills in advanced networking technologies and AI, with hands-on experience in traditional and cutting-edge practices.
- **Professional Growth:** Actively pursuing certifications in cloud computing, DevOps, and technologies like large language models.
- **Collaboration & Innovation:** Passionate about using modern tech stacks to solve real-world problems and drive business transformation.